

Noxed — Adaptive Learning for Egyptian School Science

Research Questions, Student-Case Taxonomy, Behavioural Metrics
and a Question-Difficulty Formula

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Abstract

Noxed is a competitive quiz game that teaches **science** to Egyptian school students and, after every match, recommends *the next question / level* that will move the learner forward the most. It is built as an ensemble of four ML modules: a **Diagnostic** module (clustering → knowledge priors), a **Behavioural Diagnosis** neural network (response metadata → a behavioural profile → a treatment plan), a **Question Bandit** (LinUCB → next question), and a **Progress-Tracking** module (Bayesian Knowledge Tracing → mastery). This note states the project’s **main research question and its sub-questions**, enumerates **all student cases from the learner’s perspective** with their observable signals and treatments, lists the **log-derived metrics/insights** the system must compute, and defines a concrete, cold-start-safe **question-difficulty formula**. The Chinese-math XES3G5M dataset used during prototyping is a *proxy only*: no Egyptian-science behavioural dataset exists yet, which makes cold-start and domain transfer first-class research problems.

1 Context and Scope

Domain: Egyptian national-curriculum **science** (physics, chemistry, biology), Arabic language, roughly grades 4–12.

Product: a competitive, timed quiz *game*; every question carries rich metadata (response time, chosen distractor, question type, cognitive load).

Goal: *personalised sequencing* — recommend the next question/level per student to maximise learning gain, subject to keeping the learner inside their Zone of Proximal Development and respecting behavioural “treatment” constraints.

Data status: the pipeline was prototyped on XES3G5M (Chinese elementary math: ~5.1M interactions, 14,453 students, 7,618 questions, 865 knowledge components, mean error rate ≈ 0.20). This is a **methodology proxy**, not Noxed’s data. The real target data (Egyptian science, with behaviour labels) **does not exist yet** and must be instrumented and/or synthesised — see §7.

2 Main Research Question and Sub-Questions

Main RQ. Can we infer, from a student’s in-game response data alone — correctness patterns, solving time, and mistake structure — both (i) *what they do not know* (knowledge gaps) and (ii) *how they behave under test conditions* (stable behavioural traits), and use these together to **recommend the next question / difficulty level** that maximises the student’s learning gain?

This decomposes into one sub-question per module plus two cross-cutting ones.

2.1 RQ1 — Knowledge diagnosis (Diagnostic module)

- RQ1** From the knowledge components (KCs) present in a student’s *wrong* vs. *correct* answers, can hierarchical clustering recover their weak vs. strong topics and yield useful initial mastery priors?
- RQ1.1 Which features (KC-route embedding, tree depth, co-occurrence, error rate) make wrong-answer and correct-answer clusters *meaningful* rather than arbitrary?
- RQ1.2 How should cluster *size* and cross-cluster *multiplicity* map to a prior in $[0, 1]$ (larger/repeated wrong-answer clusters $\Rightarrow$ lower prior; the inverse for correct)?
- RQ1.3 How do we exploit the KC *prerequisite tree* so a low prior on a parent KC suppresses its children?

2.2 RQ2 — Behavioural diagnosis (Behavioural NN)

- RQ2** Can a neural network classify a student’s *behavioural scenario* (e.g. poor time management, impulsivity, test anxiety, weak context-switching) from response metadata, and how well does each latent trait separate from the others?
- RQ2.1 Which observable metrics operationalise each trait? (See §4.)
- RQ2.2 Are these traits *stable* enough (across matches/sessions) to be treated as latent student properties rather than momentary noise?
- RQ2.3 What is the *ground truth*? Absent labels, can synthetic “scenario” generators and/or weak-supervision heuristics (wheel-spinning, gaming, carelessness rules) train the NN, and does it transfer to real students?
- RQ2.4 Does near-identical behaviour reliably map to near-identical treatment plans (smoothness/consistency of the score $\rightarrow$ plan rule layer)?

2.3 RQ3 — Next-question recommendation (Question Bandit)

- RQ3** Given a student’s priors and treatment plan, can a contextual bandit (LinUCB) select the next question that maximises expected learning gain while staying inside the ZPD?
- RQ3.1 What is the right *reward*? The current form $R = \frac{1}{\sum \text{priors}} \cdot Q_{\text{diff}} \cdot Q_{\text{learn}}$ favours low-mastery, harder, high-improvement items — does it need a *desirable-difficulty* term so success probability stays near a target ($\approx 0.7\text{--}0.85$)? (See §5.5.)
- RQ3.2 What context features (student ability vector, item features, behavioural scores) should be the LinUCB arm context?
- RQ3.3 How do we *validate* a bandit policy offline (off-policy evaluation) before any live student is exposed to it?

2.4 RQ4 — Progress & mastery (Progress Tracking)

- RQ4** Does BKT-based mastery tracking, updated after each match, correctly detect when a student has mastered a level and should be advanced?
- RQ4.1 How are BKT parameters $P(L_0), P(T), P(S), P(G)$ initialised from static question features before response data exists? (See §5.4.)
- RQ4.2 Does solving a question actually reduce a student’s error on its KCs (practice effect), and by how much (learning Δ)?

2.5 RQ5 & RQ6 — Cross-cutting

Start / transfer) With no Egyptian-science behavioural data, how far can models trained on proxy datasets (XES3G5M, ASSISTments, EdNet) transfer, and what is the minimum data Noxed must collect itself?

Efficacy / ATI) Does trait-matched sequencing (*aptitude–treatment interaction*) actually improve learning versus a difficulty-only baseline, and for which student profiles?

3 All Student Cases (Learner’s Perspective)

The behavioural target space is a four-family taxonomy of student “scenarios” (authored for the Egyptian context). Each case is defined by an **observable in-game signal** (what the NN/rules detect) and a **treatment** (a constraint added to the treatment plan). These are the classes the Behavioural NN must ultimately recognise.

3.1 Family A — Content Gaps (academic mastery)

Case	Diagnosis (observable signal)	Treatment (plan constraint)
Knowledge Absence	Random guessing on ordering/matching items; very high failure on a lesson; distractors chosen with no scientific logic.	Stop complex application items; serve True/False and direct definitions to rebuild the base.
Weak Prior Knowledge	Wrong answers rooted in <i>foundational</i> errors (e.g. an old formula), with long time spent.	Insert scaffolding/prerequisite items before the target KC; give rule-reminder hints.
Rote w/o Understanding	Matches a term to its literal definition correctly, but fails to <i>apply</i> it; swaps two adjacent steps in ordering items.	Switch the same content to visual item types (Image MCQ / drag-and-drop) to bind concept to image, not text.
Misconceptions	Places elements in logic-reversed order; picks a distractor built on a common science myth.	“Cognitive-conflict” cue (distinct SFX) + refutational feedback that dismantles the wrong model.

3.2 Family B — Cognitive Skills (information processing)

Case	Diagnosis (observable signal)	Treatment (plan constraint)
Working Memory	Performance collapses on high Cognitive-Load-Index items (many variables / steps).	Fade out background music during the item to free focus; prompt “use pen and paper”.
Processing Speed	Correct but always consumes maximum time; more errors on time-pressure items.	Reduce time-pressure items early; nudge “accuracy first, then speed”.
Attention (selective/sustained)	Performance drops on the <i>last</i> items of a match; errors when a visual distractor (fog/monster) is present.	1-second silence-pause to reset attention before a distractor item; avoid linguistically complex items late in the game.
Cognitive Flexibility	Sharp drop / slower responses right after a <i>topic switch</i> (context switch cost).	Progress from blocked practice to gradual interleaving, with an advance warning of a rule change.

3.3 Family C — Exam Skills (tactical)

Case	Diagnosis (observable signal)	Treatment (plan constraint)
Question Interpretation	Picks the “obvious” wrong answer on negation/exception (<i>except, not</i>) items, and does so <i>fast</i> .	Fading: attach a distinctive cue to negation items to build attention, then gradually withdraw it.
Time Mgmt. & Perfectionism	Spends 90% of time on trivially easy items; or times out on many items.	Internal-pacer SFX at 50% time elapsed; train with short, fast items to break hesitation.

3.4 Family D — Psychological / Behavioural (affect under pressure)

Case	Diagnosis (observable signal)	Treatment (plan constraint)
Impulsiveness	Responds in $< 30\%$ of the theoretical time and picks the obvious/impulsive distractor.	Slow the music tempo; disable auto score-multipliers to force deliberation.
Stress & Overthinking	Gross errors on easy items, or doubled time, <i>only</i> under stress triggers (last place, direct threat from a rival).	Graded exposure: heartbeat-tempo music in critical moments to train focus under pressure; give a confidence-restoring direct item next round.

*Note: these treatments are game-side interventions (audio, item-type, pacing, scaffolding); the Behavioural module emits them only from a fixed **treatment vocabulary** so the bandit’s filtering is well-defined.*

4 Metrics & Insights the System Must Compute

To detect the cases above, every response is reduced to log-derived features. These are grouped by the construct they indicate; they are the candidate inputs to the Behavioural NN and the signals behind each treatment trigger.

Metric (log-derived)	Construct	Definition / signal
Normalised response time $\tau = t/t_{\text{expected}}$	Speed, impulsivity	Ratio of solving time to the item’s expected time.
Fast-error rate	Impulsivity	Fraction of wrong answers with $\tau < 0.3$.
Slow-correct rate	Processing speed	Fraction of correct answers at $\tau \geq 1$ (max time).
Timeout rate	Time management	Fraction of items with no answer before the timer.
Time-on-easy share	Perfectionism	Time spent on lowest-difficulty items / total time.
Carelessness (slip)	Attention	Errors on items whose KC mastery $P(L)$ is high.
Wheel-spinning	Persistence	$\geq N$ attempts on a KC with no rise in mastery.
Gaming-the-system	Disengagement	Systematic fast wrong answers / hint abuse.
Distractor-type profile	Misconception	Distribution over impulsive vs. misconception vs. prerequisite distractors chosen.
Negation-item error gap	Interpretation	Error on negation/“except” items minus error on plain items.
Context-switch cost	Cognitive flexibility	$\Delta(\text{error, RT})$ on the first item after a topic change vs. within-topic.
Cognitive-load sensitivity	Working memory	Slope of error vs. Cognitive-Load-Index.
End-of-match decay	Sustained attention	Error on last- k items minus first- k items.
Pressure sensitivity	Test anxiety	Error/RT change under stress game-states (last place, rival threat).
Recovery rate	Emotional regulation	Accuracy on the item following a wrong/stress event.
Learning Δ per KC	Practice effect	Error-rate drop on a KC before vs. after solving related items.
Differentiation δ	Item quality	Item error minus the student’s baseline error on its KCs.
Attempt count / KC	Persistence	Number of tries before first success on a KC.
Answer-change rate	Anxiety / overthinking	Frequency of switching a selected answer before submit (if UI allows).
Session engagement	Affect	Match completion, drop-off, re-entry latency.

Key project insights.

1. **Desirable difficulty, not maximum difficulty.** Learning is fastest when success probability sits near 0.7–0.85; the reward must target that band, not simply the hardest item (§5.5).
2. **Behaviour needs its own labels.** Correctness alone cannot label anxiety or impulsivity; ground truth must come from synthetic scenarios and/or weak-supervision rules — this is the single biggest data risk.
3. **Cold start dominates the roadmap.** With no Egyptian-science data, item difficulty and

BKT priors must be derived from *static* features first, then blended with evidence as it accrues (§5).

4. **The KC tree is the backbone.** Prerequisite structure (up to depth 7 in the proxy) constrains both what to recommend next and how errors propagate to parent topics.
5. **Practice effect is measurable and directional.** A negative before/after error delta on a KC is direct evidence the sequence is teaching — use it to validate RQ4.

Measurement caveats. (1) Almost every raw metric must be *item-difficulty-normalised* (per-item z -score / lognormal residual) before it separates *behaviour* from *ability*. (2) Under a competitive timer, test anxiety, impulsivity and rapid-guessing are mutually confounded — prefer *within-student* contrasts (same student across pressure/difficulty conditions) over raw cross-sectional values. (3) The ≈ 0.8 target success probability is a design heuristic, not a fixed constant, and ATI trait→treatment matches must be A/B-validated, not assumed (Cronbach & Snow’s own caveat).

5 Question-Difficulty Formula

Difficulty $D \in [0, 1]$ is required by the bandit reward and by the BKT priors, but at launch *no response data exists* for Egyptian-science items. We therefore define difficulty in three layers: a **static** estimate from authoring features (works at cold start), an **empirical** estimate from response data (works once logs exist), and a **confidence-weighted blend** that transitions smoothly from one to the other.

5.1 Static (cold-start) difficulty

Let each item carry expert-authored features, each normalised to $[0, 1]$:

x_1 cognitive-load index (steps, variables)	x_4 item-type base (TF < MCQ < multi < fill/DnD)
x_2 KC depth in the tree (depth/7)	x_5 linguistic challenge (negation/“except”)
x_3 KC integration ($\min(k, K)/K$)	x_6 distractor plausibility

Combine them through a logistic link so the output is *bounded in $[0, 1]$ by construction*, not by clipping:

$$D_{\text{stat}} = \sigma\left(\beta_0 + \sum_{i=1}^6 \beta_i x_i\right), \quad \sigma(z) = \frac{1}{1+e^{-z}}$$

with non-negative weights β_i (higher load / depth / integration / type / language / distractor strength $\Rightarrow$ harder). Weights are set by expert calibration first, then fit by logistic regression against D_{emp} once data arrives.

5.2 Empirical difficulty (once responses exist)

Raw error rate is unreliable for rarely-attempted items, so shrink it toward the global mean μ (Beta–Binomial / empirical Bayes) with pseudo-count m :

$$D_{\text{emp}} = \frac{w + \mu m}{a + m} \quad \begin{array}{l} a = \# \text{attempts}, w = \# \text{wrong}, \\ \mu = \text{global mean error } (\approx 0.20 \text{ in the proxy}), m \approx 20. \end{array}$$

Optionally correct for *intrinsic* hardness beyond topic mastery using the differentiation delta δ (item error minus the student’s baseline error on the item’s KCs):

$$D_{\text{emp}}^* = \text{clip}(D_{\text{emp}} + \gamma \delta, 0, 1), \quad \gamma \in [0, 1].$$

5.3 Blended difficulty (cold start $\rightarrow$ data-driven)

Weight the empirical term by how much evidence backs it, $\lambda(a) = a/(a + a_0)$ (reliability constant $a_0 \approx 30$):

$$D = \lambda(a) D_{\text{emp}}^* + (1 - \lambda(a)) D_{\text{stat}}$$

At $a = 0$ (a brand-new Egyptian-science item) $D = D_{\text{stat}}$; as responses accumulate $D \rightarrow D_{\text{emp}}^*$. Being a convex combination of two quantities in $[0, 1]$, $D \in [0, 1]$ automatically — a **property directly testable with Hypothesis** for any input.

5.4 From difficulty to BKT priors

The same D (and load x_1 , option count o , type) seeds the per-item BKT parameters, each clipped to identifiable ranges ($P(S), P(G) < 0.5$):

$$P(G) = \text{clip}(g_0 (1 - D) \cdot \frac{1}{o}, 0, 0.3), \quad P(S) = \text{clip}(s_0 (0.5 + 0.5D)(0.5 + 0.5x_1), 0, 0.3),$$

$$P(L_0) = \text{clip}(0.5(1-D), 0, 1) \text{ (fallback; the Diagnostic module overrides it)}, \quad P(T) = t_0 (1-0.5D).$$

5.5 Difficulty in the bandit reward

The given reward $R = \frac{1}{\sum_j \text{prior}_j} Q_{\text{diff}} Q_{\text{learn}}$ rewards low mastery, high difficulty and high learn-rate. To keep the learner in the ZPD, replace raw Q_{diff} with a *desirable-difficulty match* on predicted success probability $p = P(L) (1 - P(S)) + (1 - P(L)) P(G)$:

$$\text{Match} = \exp\left(-\frac{(p-p^*)^2}{2\tau^2}\right), \quad p^* \approx 0.8, \quad R' = \frac{1}{\sum_j \text{prior}_j} \text{Match} Q_{\text{learn}}.$$

This peaks when the item is challenging-but-winnable rather than merely hard.

6 Validation Plan (properties to test)

- **Partitioning:** every clustered answer belongs to exactly one cluster.
- **Ranges:** behavioural scores, $P(L)$, and every D variant $\in [0, 1]$; $\Delta\text{mastery} \in [-1, 1]$ (fuzz with Hypothesis).
- **Vocabulary:** every emitted treatment constraint is a member of the fixed treatment vocabulary.
- **Smoothness:** near-identical behavioural score vectors map to the same/very similar treatment plan.
- **Bandit safety:** the selected question is always inside the constraint-filtered candidate set.
- **Efficacy:** offline off-policy evaluation shows the policy beats a difficulty-only baseline before any live rollout (RQ6).

7 Data Strategy: Proxy, Cold Start, Synthesis

A verified survey of public datasets yields **three non-overlapping buckets**: (a) large knowledge-tracing logs with rich per-attempt metadata but *no* behavioural labels; (b) small human-observed affect/behaviour corpora (BROMP lineage) that are log-linked but tiny; (c) webcam/video affect corpora with labels but *no* response metadata. **No public dataset offers both scale and behavioural ground truth, and none matches Noxed’s target** (Arabic $\times$ school-science $\times$ per-attempt logs).

Proxy / transfer datasets (methodology only). RT = response time, Att = attempts, Hint = hint use, KC = skill tags, Aff = affect labels.

Dataset	Domain / size	RT	Att	Hint	KC	Aff
XES3G5M (current)	CN elem. math; 18,066 stu, 7,652 q, 865 KC, 5.55M int	~	–	–	✓✓	–
ASSISTments 2012-13 <i>+affect</i>	US math; 27K stu, 2.5M int	✓	✓	✓	✓	(pred.)
ASSISTments 2017	US math; clickstream, 942K rows	✓	✓	✓	✓	(pred.)
EdNet KT1–KT4 (Riiid)	EN TOEIC; 784K stu, 131M int	✓✓	✓	~	✓	–
Eedi / NeurIPS- 2020	UK math MCQ; >20M answers	–	✓	–	✓✓	–
Junyi 2020 (2015)	TW math; 72K stu, 16M att	✓	✓	✓	✓	–
KDD Cup 2010	US algebra; ~30M steps	✓	✓	✓	✓	–
Squirrel AI Elem- Math2021	CN elem. math; 125K stu, difficulty+prereqs	✓	✓	–	✓	–

Closest to Noxed’s domain: TIMSS Grade 4/8 science (IEA) — **Egypt participates**, many Arabic countries included, item-level responses public, and eTIMSS adds response-time process data. It is cross-sectional assessment data (no hints/retries/sequences), so mine it for *science-item content and difficulty realism*, and use a math KT log (EdNet / Junyi / ASSISTments) as the *format & behaviour* proxy. (PISA science is computer-based with time-on-task but Egypt does not take PISA.)

Cold-start reality. There is no Arabic, per-attempt, school-science tutoring dataset publicly available; moreover *test anxiety* and *impulsiveness* are not even standard observation categories (they come from self-report). Noxed must therefore:

1. **Instrument its own game** from day one — log response time, attempts, chosen distractor, lifeline/skip/answer-change events, and stress game-states.
2. **Synthesise labelled behavioural scenarios** — parametrised “impulsive” (short RT + lure-distractor), “anxious” (RT inflation + decay under pressure), “slow-processor”, “gamer” (implausibly short RT + wrong) students — to pre-train the Behavioural NN.
3. **Anchor weak supervision** on ASSISTments-with-affect “silver” labels and the BROMP/sensor-free-detector definitions, then validate a subset against a short Arabic-validated self-report (e.g. Cognitive Test Anxiety Scale).

Item difficulty and BKT priors come from the *static* formula (§5) until enough real responses accrue.

8 Reusable Open-Source Landscape

Knowledge tracing / mastery: pyBKT (BKT priors & mastery — already in stack), pyKT and EduKTM (DKT/DKVMN/SAKT/AKT baselines), py-fsrs/ebisu (forgetting/review timing).

Adaptive tutor reference: OATutor (open BKT-based ITS with a mastery→selection loop — closest architectural analog).

Contextual bandit (M3): contextualbandits (David Cortes — LinUCB *plus* off-policy evaluation), Vowpal Wabbit (scale), catsim/EduCAT (information-theoretic CAT as a principled benchmark).

Item calibration: py-irt, girth, mirt (difficulty/discrimination to seed the difficulty formula & priors).

Behavioural (M2): no turnkey library — reuse the *definitions* of wheel-spinning / gaming / carelessness detectors (Baker, Paquette, Beck) and pyKT auxiliary-task heads; train a custom PyTorch

classifier. (Note: “OATutor”, from the same lab as *pyBKT*, is the closest end-to-end open-source analog to Noxed’s mastery→selection loop.)

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